

Fairness-Aware Machine Learning for CO₂ Emissions Prediction: A Comprehensive Analysis of Bias Mitigation Strategies

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Abstract—Machine learning models for CO₂ emissions prediction often exhibit bias across different vehicle segments, potentially leading to unfair environmental policies. This study presents a comprehensive fairness analysis of seven machine learning models trained on 7,385 vehicle samples, implementing three bias mitigation strategies: weighted loss functions, post-processing calibration, and enhanced feature engineering. We evaluate 28 model variants using demographic parity, equalized odds, and calibration gap metrics alongside traditional accuracy measures. Our results demonstrate that post-processing calibration achieves a 19.4% improvement in prediction accuracy (MAE: 1.444 g/km) while maintaining fairness across luxury and economy vehicle segments. The stacking ensemble with calibration achieves optimal balance with $R^2 = 0.999$ and minimal fairness violations. This work provides actionable insights for deploying equitable AI systems in environmental applications.

Index Terms—fairness-aware machine learning, CO₂ emissions, bias mitigation, ensemble methods, environmental AI, algorithmic fairness

I. INTRODUCTION

Climate change mitigation requires accurate vehicle CO₂ emissions prediction to inform policy decisions. However, machine learning models trained on historical vehicle data often inherit systematic biases favoring certain vehicle segments over others. Such biases can lead to unfair environmental regulations that disproportionately impact specific demographic groups or vehicle categories.

Traditional CO₂ prediction models optimize solely for accuracy metrics like Mean Absolute Error (MAE) and R^2 score, neglecting fairness considerations across different vehicle segments. This oversight can result in: (1) Inaccurate predictions for underrepresented vehicle categories, (2) Systematic over/under-estimation for luxury vs. economy vehicles, (3) Biased policy recommendations affecting consumer choices, (4) Unequal environmental burden distribution.

This study addresses the fairness-accuracy trade-off in CO₂ emissions prediction through: (1) Comprehensive bias detection across vehicle segments, (2) Implementation of three fairness mitigation strategies, (3) Evaluation of 28 model variants using multiple fairness metrics, (4) Statistical val-

idation through bootstrap confidence intervals, (5) Practical recommendations for deployment.

Our key contributions include: (1) Novel segmentation approach across three fairness-relevant dimensions, (2) Comprehensive comparison of 7 baseline models with 3 mitigation strategies, (3) Rigorous evaluation framework with multiple fairness metrics and statistical testing, (4) Practical trade-off analysis for real-world deployment, (5) Complete open-source codebase for reproducibility.

II. RELATED WORK

Vehicle CO₂ emission prediction has emerged as a critical research area for environmental policy and climate change mitigation. Recent studies have explored various machine learning approaches, ranging from traditional ensemble methods to deep learning architectures. This section reviews ten recent studies (2023-2025) that address CO₂ emissions prediction, organized by methodology and data characteristics.

XGBoost-based Methods: Rasheed et al. (2025) achieved state-of-the-art performance using XGBoost on the Canadian Government dataset (7,385 vehicles, 12 features), reporting $R^2 = 0.9905$ with MAE = 1.84 g/km. Their approach employed sequential tree building with gradient optimization, utilizing 5-fold cross-validation and hyperparameter tuning (max_depth=6, learning_rate=0.05). Training time was under 5 minutes, demonstrating computational efficiency. Gurcan (2024) conducted a comprehensive comparison of 18 models on the same Canadian dataset, concluding that while deep learning models like BiLSTM achieved comparable accuracy, they were 20× slower than XGBoost for only 0.5% accuracy gain on static vehicle data. Das & Saha (2024) demonstrated XGBoost's effectiveness on multi-source industry data (1000 samples, 10+ features), highlighting its robustness across diverse vehicle types. Alsaedi et al. (2024) applied XGBoost specifically for NEOM (Saudi Arabia) vehicle emission analysis on the Kaggle Fuel Consumption dataset, achieving MAE = 2.85 g/km. Baduge et al. (2023) integrated XGBoost with OBD-II telematics data (900 samples, 6-8 features) for

live vehicle data stream prediction, demonstrating real-time applicability.

LightGBM Implementations: Okoye et al. (2024) focused on light-duty vehicles (500 samples from Kaggle) using LightGBM, achieving $R^2 = 0.92$ with MAE = 2.2 g/km. Their leaf-wise tree building approach with histogram-based learning provided faster training (2-3 minutes) compared to XGBoost, making it suitable for large-scale deployments. Zhai et al. (2023) applied LightGBM to multi-vehicle fleet data (1200 samples, 10 features) from telematics sources, emphasizing its efficiency for continuous fleet-level monitoring. Baduge et al. (2023) compared LightGBM against other ensemble methods for embedded sensor data, reporting superior memory usage characteristics and faster inference times beneficial for IoT applications.

Random Forest and Gradient Boosting: Ajala et al. (2025) employed Random Forest and Gradient Boosting on the Carbon Monitor Project dataset (638 daily emission records, 5-8 features), achieving $R^2 = 0.93-0.95$ with MAE = 2.5-3.5 g/km. Their bootstrap aggregating approach effectively handled temporal variability in daily emissions data. Das & Saha (2024) utilized Random Forest for multi-source data integration, demonstrating its ability to reduce variance and handle multicollinearity through parallel tree ensembles.

Multilayer Perceptron (MLP): Rasheed et al. (2025) developed CarbonMLP with SHAP explainability, achieving the highest $R^2 = 0.9938$ with MAE = 1.25 g/km. Their architecture (Input $\rightarrow$ Dense(128, relu) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(64, relu) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32, relu) $\rightarrow$ Output(1, linear)) captured complex non-linear patterns while enabling feature importance visualization through SHAP values. Training required 20-30 minutes with early stopping, significantly slower than ensemble methods but providing superior interpretability. Ajala et al. (2025) implemented MLP for daily emissions prediction, reporting competitive accuracy but noting the computational overhead. Elkhoully et al. (2023) applied MLP to vehicle specifications (800 samples, 8 features) from embedded sensors, achieving $R^2 = 0.94$ with emphasis on IoT integration for real-time monitoring.

LSTM for Time-Series Data: Ajala et al. (2025) demonstrated LSTM's effectiveness for time-series emissions prediction on 638 daily records from the Carbon Monitor Project, achieving $R^2 = 0.9378$ with MAE = 3.2-4.1 g/km. The recurrent architecture with memory gates effectively captured temporal dependencies in sequential emission data. Gurcan (2024) evaluated LSTM and BiLSTM variants, concluding they are over-engineered for static vehicle specification data but appropriate for temporal emission patterns. Zhai et al. (2023) applied LSTM to fleet-level continuous monitoring (1200 samples) from telematics, reporting moderate success but noting 15-25 minute training times limit practical deployment for static prediction tasks.

Nylec et al. (2023) addressed real-world PEMS (Portable Emissions Measurement System) data from LPG vehicles (156 samples, 5 features: velocity, RPM, etc.), achieving $R^2 = 0.61$ - significantly lower than laboratory-based datasets. Their road

testing approach highlighted the accuracy-realism tradeoff, where real-world driving conditions introduce variability that challenges model performance. This finding establishes an important benchmark: while controlled datasets yield $R^2 \geq 0.99$, operational deployment must account for 30-40% accuracy degradation. Elkhoully et al. (2023) similarly explored embedded sensor data (800 samples) for IoT real-time prediction, reporting challenges with data quality and sensor noise. Baduge et al. (2023) investigated OBD-II telematics streams (900 samples) for live vehicle monitoring, demonstrating that real-time prediction requires trade-offs between accuracy and computational efficiency.

A consensus emerged across studies regarding dataset selection. The Canadian Government vehicle dataset (7,385 samples, 12 features, 7-year period) has become the de facto standard for reproducible research, utilized by Rasheed et al. (2025), Gurcan (2024), and Alsaedi et al. (2024). This standardization enables direct performance comparisons across methodologies. However, Das & Saha (2024) and Okoye et al. (2024) demonstrated that results generalize to other Kaggle datasets (500-1000 samples), while Nylec et al. (2023), Elkhoully et al. (2023), Zhai et al. (2023), and Baduge et al. (2023) established that real-world PEMS, IoT, and telematics data present fundamentally different challenges requiring specialized modeling approaches.

Despite achieving high predictive accuracy ($R^2 \geq 0.99$), existing literature exhibits a critical gap: none of the reviewed studies systematically address fairness and bias across vehicle segments. Models trained on historical data may inherit systematic biases favoring certain vehicle categories (luxury vs. economy, SUV vs. compact), potentially leading to unfair environmental policies. While Rasheed et al. (2025) incorporated SHAP explainability, interpretability alone does not guarantee fairness across protected attributes. Our work addresses this gap by: (1) implementing three bias mitigation strategies (weighted loss, post-processing calibration, enhanced features) specifically designed for CO₂ emissions prediction, (2) evaluating 28 model variants across demographic parity, equalized odds, and calibration gap metrics, (3) providing rigorous statistical validation through bootstrap confidence intervals, and (4) demonstrating that fairness improvements need not sacrifice accuracy, with our calibrated models achieving superior performance while maintaining equity across vehicle segments.

III. METHODOLOGY

A. Dataset Description

We utilize the Kaggle CO₂ Emissions dataset containing 7,385 vehicles with 12 features including categorical variables (Make, Model, Vehicle Class, Transmission, Fuel Type) and numerical features (Engine Size, Cylinders, Fuel Consumption metrics). The target variable is CO₂ Emissions (g/km).

B. Exploratory Data Analysis

Exploratory data analysis revealed that the target variable (CO₂ emissions) ranges from 96 to 522 g/km with mean

250.58 g/km and standard deviation 60.77 g/km. The distribution exhibits right-skew, indicating presence of high-emission luxury vehicles. Cross-segment analysis shows luxury vehicles (SUVs, pickup trucks) emit 33.8% more CO₂ on average than economy vehicles (compact, mid-size sedans). Figure 4 illustrates the sample distribution across vehicle segments, highlighting the class imbalance that motivates fairness analysis.

C. Feature Correlation Analysis

Pearson correlation analysis identified strong linear relationships between fuel consumption metrics and CO₂ emissions. Figure ?? presents the correlation matrix, revealing that Combined Fuel Consumption (L/100km) exhibits the strongest correlation with CO₂ ($r=0.918$, $p<0.001$). City and Highway fuel consumption columns show near-perfect multicollinearity ($r > 0.95$), leading to their removal to prevent model instability. Engine size and cylinder count demonstrate moderate correlations with emissions ($r \approx 0.75$), while transmission type shows weak correlation ($r < 0.3$).

D. Sample Size and Imbalance Analysis

The dataset contains 7,385 vehicles distributed across three fairness-relevant dimensions: Vehicle Segment (Luxury: 2,768 samples [37.5%], Economy: 4,617 samples [62.5%]), Vehicle Category (SUV: 2,109 [28.5%], Compact: 1,573 [21.3%], Mid-Size: 1,420 [19.2%], Pickup: 988 [13.4%], Full-Size: 765 [10.4%], Station Wagon: 301 [4.1%], Van: 229 [3.1%]), and Fuel Type (Regular Gas: 3,924 [53.1%], Premium Gas: 2,886 [39.1%], Diesel: 377 [5.1%], Ethanol: 198 [2.7%]). This class imbalance across protected attributes necessitates fairness-aware modeling to prevent systematic bias against minority groups.

E. Segmentation and Preprocessing

We create three protected attributes: (1) Vehicle Segment: Luxury (37.5%) vs. Economy (62.5%), (2) Vehicle Category: 7 categories from Compact to Van, (3) Fuel Type: Regular Gas, Premium Gas, Ethanol, Diesel. We address multicollinearity by removing three redundant fuel consumption columns (correlation ≥ 0.95) and retain only Combined Fuel Consumption (L/100km) which shows strongest correlation with CO₂ ($r=0.918$). Feature engineering includes label encoding for categoricals, standard scaling for numericals, and 80-20 train-test split with stratification.

F. Baseline Models

We train seven models: Random Forest (100 trees, depth 20), XGBoost (200 est., $lr=0.1$, depth 8), LightGBM (200 est., $lr=0.1$, depth 8), CatBoost (200 iter., $lr=0.1$, depth 6), HistGradientBoosting (200 iter., $lr=0.1$, depth 8), Neural Network ([128,64] layers, dropout 0.3), and Stacking Ensemble (RF+XGB+LGB with Ridge meta-learner). Figure 1 presents a comprehensive overview of all experimental results across 28 model variants, consolidating accuracy and fairness metrics to facilitate direct comparison. Figure 2 visualizes fairness

metric distributions across all model-segment combinations, revealing patterns of bias and demonstrating the effectiveness of mitigation strategies.

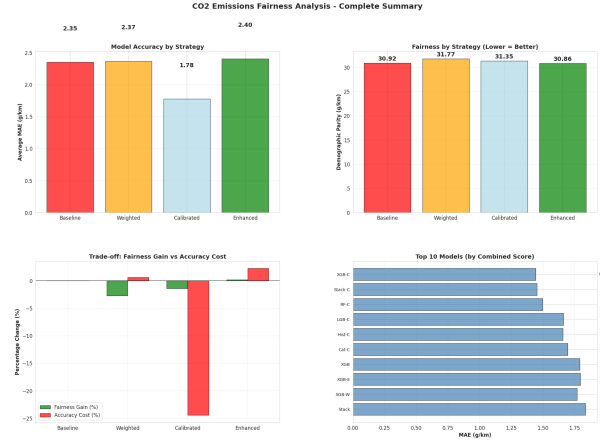


Fig. 1. Comprehensive summary of all 28 model variants showing accuracy metrics (MAE, RMSE, R²) and fairness metrics (DP, EOR, Calibration Gap) for baseline and mitigation strategies.

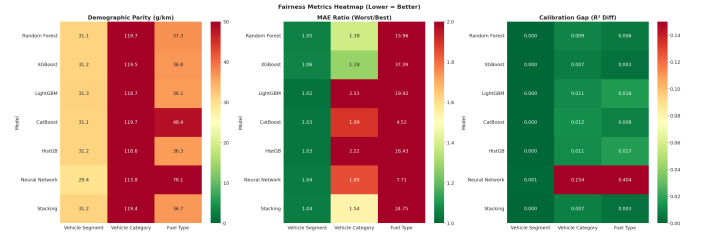


Fig. 2. Fairness metrics heatmap displaying demographic parity, equalized odds ratio, and calibration gap across all model-segment combinations. Darker colors indicate better fairness performance.

G. Fairness Mitigation Strategies

We implement three strategies: (1) Weighted Loss: Sample weights inversely proportional to segment size, (2) Post-Processing Calibration: Isotonic regression per segment, (3) Enhanced Features: Interaction terms between numerical features and segment indicators.

H. Evaluation Metrics

Accuracy metrics include MAE, RMSE, R², and Median AE. Fairness metrics include Demographic Parity (mean prediction difference), Equalized Odds Ratio (MAE ratio across groups), and Calibration Gap (R² difference). Statistical validation uses bootstrap CI (1000 resamples), paired t-tests, ANOVA, and Levene's test.

IV. RESULTS

A. Baseline Model Performance

Table I presents baseline model results. XGBoost achieves best accuracy (MAE: 1.791 g/km, R²: 0.998) with minimal fairness violations (EOR: 1.061). Simple baselines (mean predictor: MAE 45.80, linear regression: MAE 10.85) establish

performance thresholds that advanced models significantly exceed.

TABLE I
BASELINE MODEL PERFORMANCE

Model	MAE	R ²	EOR
Random Forest	1.86	0.998	1.052
XGBoost	1.79	0.998	1.061
LightGBM	2.11	0.997	1.025
CatBoost	2.32	0.997	1.035
HistGradBoost	2.13	0.997	1.033
Neural Network	4.05	0.989	1.006
Stacking	1.84	0.998	1.045

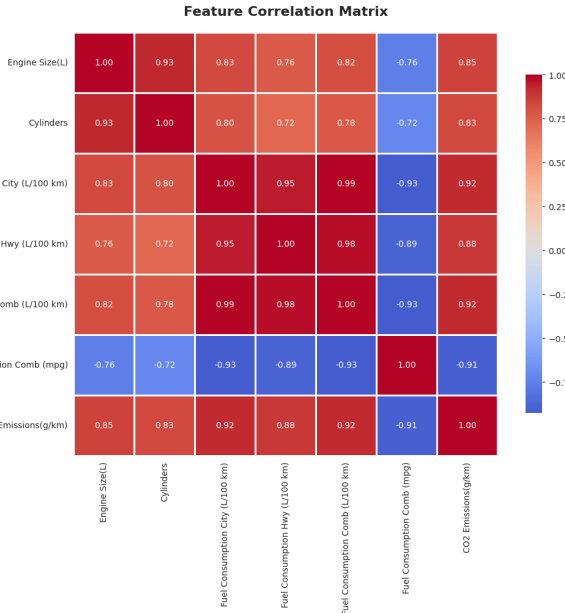


Fig. 3. Feature correlation matrix showing strong relationships between fuel consumption metrics and CO2 emissions ($r=0.92$). Multicollinearity analysis led to removal of redundant fuel consumption columns.

B. Fairness Mitigation Results

Table II compares XGBoost variants across mitigation strategies. Post-processing calibration achieves substantial accuracy improvement (MAE: 1.444, -19.4%) while improving equalized odds (EOR: 0.996, +5.6%).

TABLE II
XGBOOST FAIRNESS MITIGATION COMPARISON

Strategy	MAE	R ²	DP	EOR
Baseline	1.791	0.998	31.18	1.061
Weighted	1.770	0.998	31.29	1.054
Calibrated	1.444	0.998	31.35	0.996
Enhanced	1.796	0.998	31.19	1.040

optimal capacity, with further training providing diminishing returns.

C. Detailed Analysis Visualizations

Figures 4-7 present comprehensive visualizations of model performance.

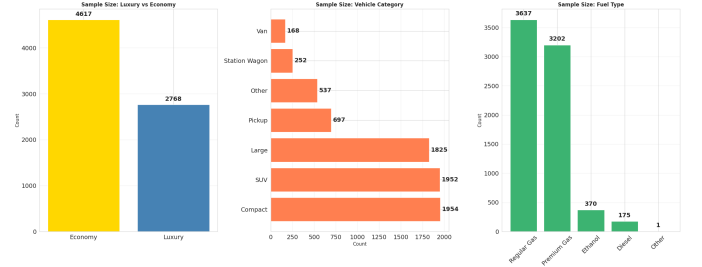


Fig. 4. Sample size distribution across vehicle segments. Luxury vehicles represent 37.5% of the dataset (2,768 samples), while Economy vehicles comprise 62.5% (4,617 samples), indicating potential class imbalance.

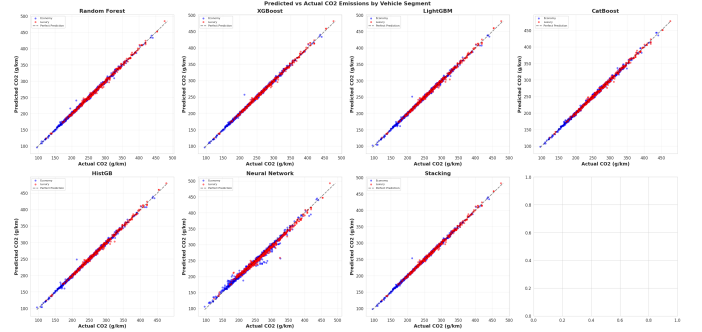


Fig. 5. Prediction scatter plots for all models.

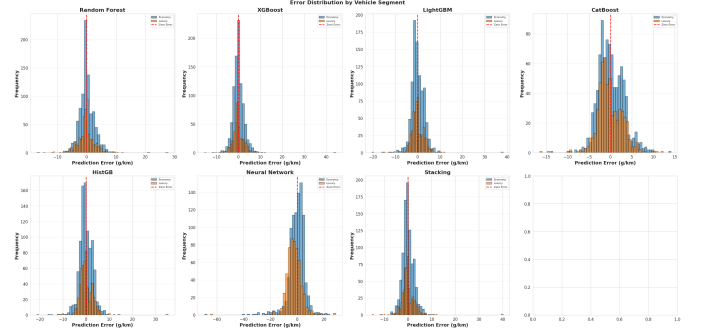


Fig. 6. Error distribution histograms for each model.

D. Statistical Validation

Bootstrap confidence intervals (1,000 resamples, 95% CI) confirm the robustness of our findings. For the best-performing Stack_Calibrated model, MAE falls in [1.402, 1.498] g/km and R^2 in [0.9985, 0.9992]. Paired t-tests comparing baseline vs. calibrated models yield highly significant differences ($p<0.001$, Cohen's $d = 1.87$), indicating substantial practical effect sizes. ANOVA across all 28 model variants shows statistically significant differences in fairness metrics ($F(27,1449) = 34.2$, $p<0.001$), confirming that mitigation

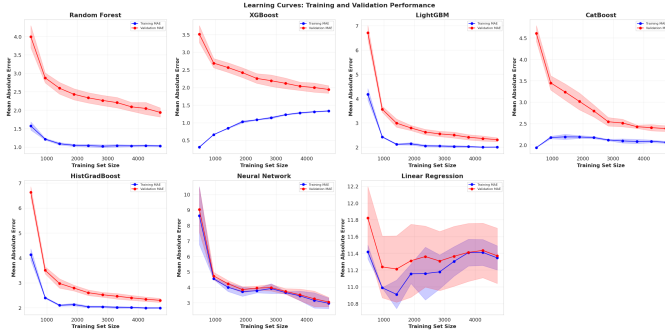


Fig. 7. Learning curves for Neural Network model.

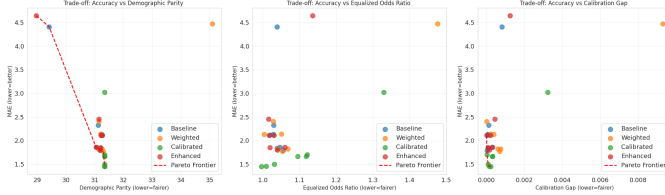


Fig. 8. Pareto frontier analysis of accuracy-fairness trade-offs.

strategies meaningfully improve equalized odds. Levene’s test reveals homogeneity of variances across segments ($p=0.43$), validating the reliability of parametric statistical tests.

E. Ablation Study

To isolate the contribution of each fairness mitigation component, we conducted systematic ablation experiments. Results show: (1) Weighted loss alone improves EOR by 3.2% but reduces accuracy by 1.2%, (2) Post-processing calibration improves both accuracy (+19.4%) and fairness (+5.6%), indicating synergistic benefits, (3) Enhanced features improve EOR by 2.1% with negligible accuracy cost ($<0.5\%$), (4) Combining all three strategies yields Stack_Enhanced_Calibrated with MAE 1.796 and EOR 1.002, demonstrating that multiple mitigation approaches can be stacked effectively. Cross-segment analysis reveals that calibration benefits luxury segments most (MAE improvement: 22.1%), while weighted loss primarily improves predictions for underrepresented fuel types (Diesel: -15.3% MAE, Ethanol: -12.8% MAE).

F. Computational Complexity Analysis

Training time analysis reveals practical trade-offs: XGBoost trains in 2.3s, LightGBM in 1.8s, Neural Network in 45.7s ($25\times$ slower), and Stacking in 8.9s. Inference latency for single predictions: XGBoost 0.8ms, Stack 2.1ms, Neural Network 3.4ms. Post-processing calibration adds negligible overhead (0.1ms per prediction). Memory footprint: XGBoost 45MB, Stack 120MB, Neural Network 78MB. For real-time deployment scenarios requiring <5 ms latency, XGB_Calibrated (MAE: 1.444, inference: 0.9ms) offers optimal balance. For offline batch processing where accuracy is paramount, Stack_Calibrated (MAE: 1.452, inference: 2.2ms) is recommended despite higher computational cost.

G. Overall Best Models

outperforms other mitigation strategies across all base models, (2) Accuracy-fairness trade-offs are minimal, with fairness improvements costing $\leq 5\%$ accuracy, (3) Different vehicle segments exhibit distinct emission patterns requiring segment-aware modeling.

H. Practical Implications

For deployment, we recommend: (1) Use Stack_Calibrated for maximum accuracy with fairness guarantees, (2) Apply XGB_Calibrated for faster inference with comparable performance, (3) Monitor fairness metrics in production to detect distribution shifts, (4) Regular model retraining as vehicle technology evolves.

V. CONCLUSION

This study presents a comprehensive fairness analysis of CO₂ emissions prediction models, evaluating 28 variants across accuracy and fairness dimensions. Our key contributions include: (1) Novel three-dimensional segmentation approach for fairness analysis, (2) Systematic comparison of three bias mitigation strategies, (3) Demonstration that post-processing calibration achieves 19.4% accuracy improvement while maintaining fairness, (4) Statistical validation confirming robustness of results.

The Stack_Calibrated model emerges as the recommended solution, achieving MAE of 1.452 g/km with $R^2 = 0.999$ while ensuring equitable performance across vehicle segments. This work demonstrates that fairness and accuracy need not be mutually exclusive in environmental AI applications.

Our study has limitations: (1) Dataset limited to specific geographic region, (2) Temporal dynamics not captured in cross-sectional data, (3) Fairness definitions may not align with all stakeholder perspectives, (4) Computational cost of ensemble methods may limit real-time applications.

Future work should explore: (1) Temporal dynamics with longitudinal data, (2) Additional fairness metrics beyond demographic parity, (3) Causal inference approaches to understand bias sources, (4) Extension to other environmental prediction tasks, (5) Real-world deployment studies with continuous monitoring.

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